Project Charter



ACE

CYBER R&D Program

Conversational AI Interface for
Analyst-Driven Threat Investigation

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Project Charter

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1.0 Problem Statement

The PI chose to research the ACE proposed 'AI-Augmented Security Operations' topic. This topic is important because analysts are currently slowed by manual log investigation processes within the Security Operations Center (SOC) environment. There is a critical need for a more intuitive, natural language interface to rapidly and efficiently query and analyze vast datasets, particularly from Elasticsearch Enterprise, to accelerate threat investigation and hunting.

2.0 Research Question

How can a Conversational AI interface specifically integrated with Elasticsearch be designed to enable SOC analysts to query log data using natural language for common adversarial threat-hunting tasks (e.g., failed logins, suspicious IPs, lateral movements)?

3.0 Hypothesis

If a Conversational AI interface is developed and integrated with Elasticsearch to support natural-language threat-hunting queries, then, when evaluated against a pre-tagged dataset containing failed-login and lateral-movement events, it will achieve an overall F1 score of at least 0.95 across all evaluated threat-hunting tasks, indicating balanced precision and recall in detecting threat-tagged log events.

4.0 Project Objectives

1. Build a functional Proof of Concept (PoC) Conversational AI interface that supports natural language queries for SOC log data.
2. Integrate with Elasticsearch to retrieve and display relevant log data.
3. Support at least two common threat-hunting tasks (e.g., failed logins, suspicious IPs).

4. Measure query latency, detection accuracy, and analyst satisfaction compared to manual methods.

5.0 Research Method Selection

Applied Experiment: introducing and testing a new Conversational AI interface system in a simulated cybersecurity context. The goal is to solve a practical problem and evaluate its impact on analyst performance, rather than generating purely theoretical knowledge.

6.0 Scope & Deliverables

Deliverables:

1. Conversational AI Prototype (Proof of Concept)
2. Query-to-Elasticsearch integration specifications
3. Evaluation of results (detection accuracy, F1 score)

Out-of-Scope Items:

1. Full production deployment
2. Comprehensive SOC automation
3. Handling real classified data or CUI (anonymized/simulated logs instead)

7.0 Resources & Tools Needed

Access to a SOC simulated Elasticsearch Enterprise environment, relevant log data (anonymized/synthesized), and acc10ess to an enclave for development/testing.

8.0 Timeline (3-Month Scope)

Month 1: Literature review, finalize research design, submit outline.
Month 2: Implement POC, test with simulated logs, submit Midpoint Status Report.Month 3: Collect and analyze results, refine evaluation, draft final report, prepare 15–20 minute presentation, conduct internal peer review, and finalize.

9.0 Success Criteria & Metrics

Success will be measured by its accuracy: Suspicious logs in the synthetic dataset will be pre-tagged and stored in a reference list accessible only to the PI. During testing, the AI interface will be evaluated against this list, with a target accuracy above 95%.

10.0 Risks & Mitigations

1. Synthetic logs may not fully reflect SOC complexity: Use varied log types to increase realism.
2. AI misinterpretation of data and hallucinations: Limit scope to 3 threat-hunting tasks with clear expected outputs.

11.0 Expected Impact

This study contributes to DoD cybersecurity research by exploring how a Conversational AI interface can streamline threat investigation in SOCs, reducing reliance on time-intensive and tedious manual queries. It enables analysts to interact with large and complex log data through natural language, enhancing speed and accuracy in detecting and responding to threats, with the ultimate goal of strengthening the DoD's cyber defense capabilities.

12.0 Change Log

Ver.	Date	Description of Change	Author
v1.1	2025-09-03	Filled out document (first time)	Koi McFarland
v1.2	2025-09-27	Changed Hypothesis	Koi McFarland
v1.3	2025-11-15	Changed Hypothesis	Koi McFarland
v1.4	2025-11-19	Updated Document Formatting	Koi McFarland
v1.5	2025-11-29	Slightly changed hypothesis	Koi McFarland

13.0 Mentor Review / Approval

Reviewer	Role	Date	Notes
Student Peer	PI	2025-12-17	Final Deliverable
Faculty Advisor			
DoD/ACE Mentor			